

```
loke4884@linuxforge:~$ mkdir -p ~/kvm-vms/linuxserver2
loke4884@linuxforge:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos kvm-vms
phoneintel-venv snap
loke4884@linuxforge:~$ cd ~/kvm-vms/linuxserver2
loke4884@linuxforge:~/kvm-vms/linuxserver2$ wget
https://cloud-images.ubuntu.com/noble/current/noble-server-cloudimg-amd64.img
--2026-09-07 11:06:45--
https://cloud-images.ubuntu.com/noble/current/noble-server-cloudimg-amd64.img
Resolving cloud-images.ubuntu.com (cloud-images.ubuntu.com)... 2620:2d:4000:1::1a,
2620:2d:4000:1::17, 185.125.190.40, ...
Connecting to cloud-images.ubuntu.com (cloud-images.ubuntu.com)[2620:2d:4000:1::1a]:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 624829952 (596M) [application/octet-stream]
Saving to: 'noble-server-cloudimg-amd64.img'
```

```
noble-server-cloudimg-amd64.img
100%[=====
=====>] 595.88M
3.10MB/s  in 3m 57s
```

```
2026-09-07 11:10:44 (2.51 MB/s) - 'noble-server-cloudimg-amd64.img' saved
[624829952/624829952]
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ ls
noble-server-cloudimg-amd64.img
loke4884@linuxforge:~/kvm-vms/linuxserver2$ qemu-img info
noble-server-cloudimg-amd64.img
image: noble-server-cloudimg-amd64.img
file format: qcow2
virtual size: 3.5 GiB (3758096384 bytes)
disk size: 596 MiB
cluster_size: 65536
Format specific information:
  compat: 1.1
  compression type: zlib
  lazy refcounts: false
  refcount bits: 16
  corrupt: false
  extended l2: false
Child node 'file':
  filename: noble-server-cloudimg-amd64.img
  protocol type: file
```

file length: 596 MiB (624829952 bytes)

disk size: 596 MiB

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ mv noble-server-cloudimg-amd64.img  
ubuntu.qcow2
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ nano user-data
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cat user-data  
#cloud-config
```

hostname: linuxserver2

manage_etc_hosts: true

users:

- name: lokesh

groups:

- sudo

sudo: ALL=(ALL) NOPASSWD:ALL

shell: /bin/bash

lock_passwd: false

passwd: \$6\$YOUR_HASH_HERE

ssh_pwauth: true

package_update: true

packages:

- openssh-server

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ openssl passwd -6
```

Password:

Verifying - Password:

\$6\$W7e9xqJuQl6ETa/G\$ACU3Mgw9Ygf8wJBWzROdP5fQcS1i0G/Lma6bfFLcp2BZqFTboP8S
CQaxSh.r6UYN0drIlyF8bLIU2HPOZYsYS1

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ nano user-data
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cat user-data  
#cloud-config
```

hostname: linuxserver2

manage_etc_hosts: true

users:

- name: owner

groups:

- sudo

sudo: ALL=(ALL) NOPASSWD:ALL

shell: /bin/bash

```
lock_passwd: false
passwd: $6$YOUR_HASH_HERE
```

```
ssh_pwauth: true
```

```
package_update: true
```

```
packages:
```

```
- openssh-server
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ nano user-data
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cat user-data
```

```
#cloud-config
```

```
hostname: linuxserver2
```

```
manage_etc_hosts: true
```

```
users:
```

```
- name: owner
```

```
groups:
```

```
- sudo
```

```
sudo: ALL=(ALL) NOPASSWD:ALL
```

```
shell: /bin/bash
```

```
lock_passwd: false
```

```
passwd:
```

```
$6$W7e9xqJuQl6ETa/G$ACU3Mgw9Ygf8wJBWzROdP5fQcS1i0G/Lma6bfFLcp2BZqFTboP8S  
CQaxSh.r6UYN0drllYF8bLIU2HPOZYsYS1
```

```
ssh_pwauth: true
```

```
package_update: true
```

```
packages:
```

```
- openssh-server
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ nano meta-data
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cat meta-data
```

```
instance-id: linuxserver2
```

```
local-hostname: linuxserver2
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cloud-localds seed.iso user-data meta-data
```

```
Command 'cloud-localds' not found, but can be installed with:
```

```
sudo apt install cloud-image-utils
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ sudo apt install cloud-image-utils -y
[sudo: authenticate] Password:
Installing:
  cloud-image-utils

.
.
.
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 genisoimage amd64 9:1.1.11-5
[293 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 cloud-image-utils all
0.33-1build1 [18.7 kB]
Fetched 312 kB in 4s (74.3 kB/s)
.
.
Processing triggers for man-db (2.13.1-1build1)...
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ sudo apt install libvirt-daemon-system virtinst -y
libvirt-daemon-system is already the newest version (12.0.0-1ubuntu5.3).
Installing:
  virtinst
```

```
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 121
  Download size: 1,964 B
  Space needed: 28.7 kB / 43.0 GB available
```

```
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 virtinst all 1:5.1.0-1 [1,964 B]
Fetched 1,964 B in 1s (3,502 B/s)
.
.
Setting up virtinst (1:5.1.0-1)...
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ ls -l /dev/kvm
crw-rw----+ 1 root kvm 10, 232 Sep  7 10:59 /dev/kvm
loke4884@linuxforge:~/kvm-vms/linuxserver2$ cloud-localds seed.iso user-data meta-data
loke4884@linuxforge:~/kvm-vms/linuxserver2$ ls
meta-data  seed.iso  ubuntu.qcow2  user-data
loke4884@linuxforge:~/kvm-vms/linuxserver2$ ls -lh
total 597M
-rw-rw-r-- 1 loke4884 loke4884  55 Sep  7 11:29 meta-data
-rw-rw-r-- 1 loke4884 loke4884 366K Sep  7 11:33 seed.iso
-rw-rw-r-- 1 loke4884 loke4884 596M Aug 26 18:50 ubuntu.qcow2
-rw-rw-r-- 1 loke4884 loke4884  376 Sep  7 11:28 user-data
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ qemu-img resize ubuntu.qcow2 20G
Image resized.
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ qemu-img info ubuntu.qcow2
```

```
image: ubuntu.qcow2
```

```
file format: qcow2
```

```
virtual size: 20 GiB (21474836480 bytes)
```

```
disk size: 596 MiB
```

```
cluster_size: 65536
```

```
Format specific information:
```

```
  compat: 1.1
```

```
  compression type: zlib
```

```
  lazy refcounts: false
```

```
  refcount bits: 16
```

```
  corrupt: false
```

```
  extended l2: false
```

```
Child node '/file':
```

```
  filename: ubuntu.qcow2
```

```
  protocol type: file
```

```
  file length: 596 MiB (624886272 bytes)
```

```
  disk size: 596 MiB
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ sudo virt-install \
```

```
--name linuxserver2 \
```

```
--memory 4096 \
```

```
--vcpus 2 \
```

```
--disk path=$PWD/ubuntu.qcow2,format=qcow2 \
```

```
--disk path=$PWD/seed.iso,device=cdrom \
```

```
--os-variant ubuntu24.04 \
```

```
--network network=default \
```

```
--graphics none \
```

```
--console pty,target_type=serial \
```

```
--import
```

```
Starting install...
```

```
Creating domain...
```

```
|    00:00:00
```

```
.
```

```
.
```

```
linuxserver2 login: owner
```

```
Password:
```

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)
```

```
* Documentation: https://help.ubuntu.com
```

```
* Management:   https://landscape.canonical.com
```

```
* Support:      https://ubuntu.com/pro
```

System information as of Mon Sep 7 06:08:54 UTC 2026

System load: 0.03 Processes: 138
Usage of /: 10.0% of 18.33GB Users logged in: 0
Memory usage: 5% IPv4 address for enp1s0: 192.168.122.31
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

68 updates can be applied immediately.
68 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

```
owner@linuxserver2:~$ hostname
linuxserver2
owner@linuxserver2:~$ hostnamectl
Static hostname: linuxserver2
    Icon name: computer-vm
    Chassis: vm
    Machine ID: 2488c98995f14b4da32fb9d441a207d2
    Boot ID: 21e3f56e5754492da13bfaac6b80ba3d
Virtualization: kvm
Operating System: Ubuntu 24.04.4 LTS
    Kernel: Linux 6.8.0-138-generic
    Architecture: x86-64
Hardware Vendor: QEMU
    Hardware Model: Standard PC _Q35 + ICH9, 2009_
Firmware Version: 1.17.0-debian-1.17.0-1ubuntu1
```

Firmware Date: Tue 2014-04-01

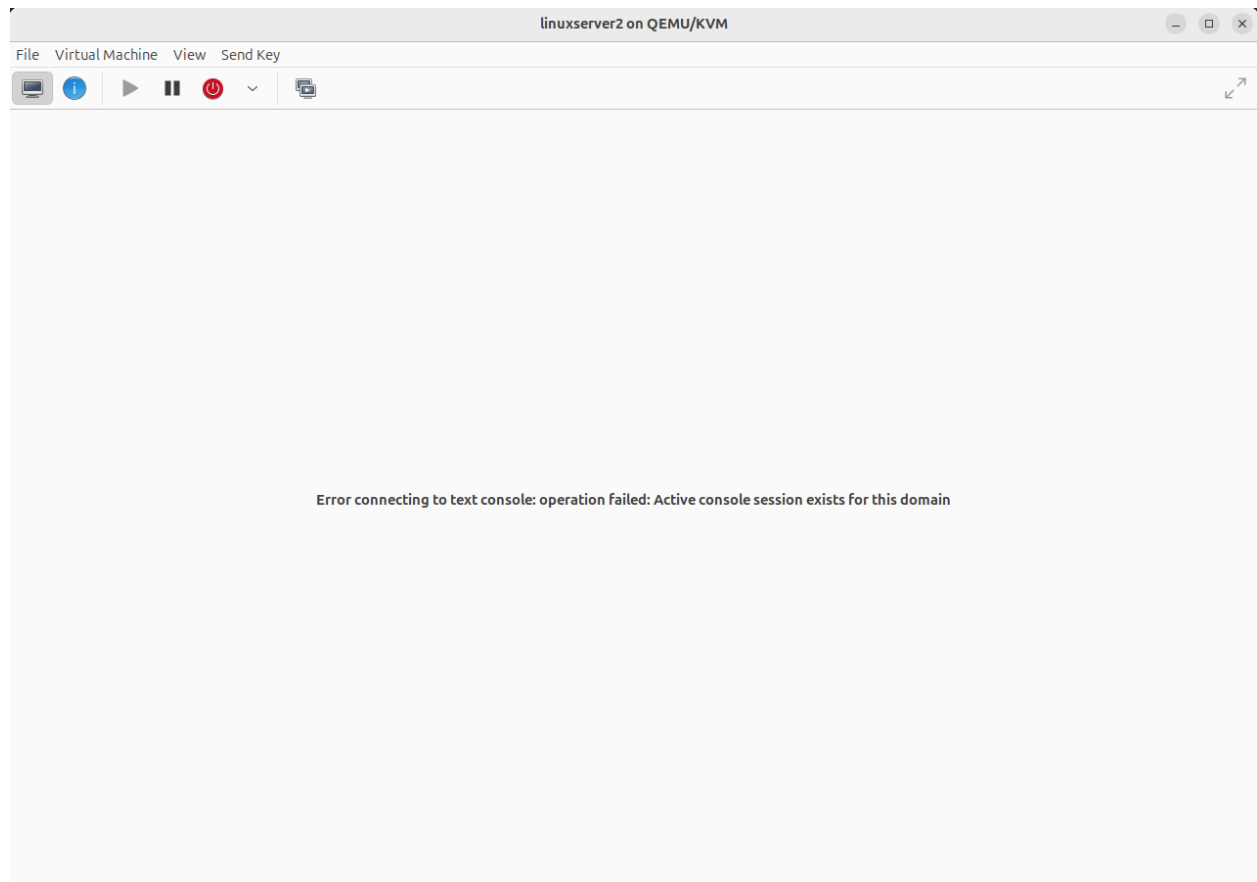
Firmware Age: 12y 5month 1w

owner@linuxserver2:~\$ ip r

default via 192.168.122.1 dev enp1s0 proto dhcp src 192.168.122.31 metric 100

192.168.122.0/24 dev enp1s0 proto kernel scope link src 192.168.122.31 metric 100

192.168.122.1 dev enp1s0 proto dhcp scope link src 192.168.122.31 metric 100



owner@linuxserver2:~\$ exit

logout

Ubuntu 24.04.4 LTS linuxserver2 ttyS0

linuxserver2 login:

Domain creation completed. (ctrl +])

```
linuxserver2 on QEMU/KVM
File Virtual Machine View Send Key
linuxserver2 login: owner
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  7 06:22:27 UTC 2026

System load:  0.0          Processes:            137
Usage of /:   10.1% of 18.3GB Users logged in:      1
Memory usage: 5%          IPv4 address for enp1s0: 192.168.122.31
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

68 updates can be applied immediately.
68 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

owner@linuxserver2:~$
```

From the hostmachine side :

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ sudo virsh list
```

[sudo: authenticate] Password:

```
Id  Name      State
-----
1   linuxserver2  running
```

```
loke4884@linuxforge:~/kvm-vms/linuxserver2$ sudo virsh dominfo linuxserver2
```

```
Id:          1
Name:        linuxserver2
UUID:        2488c989-95f1-4b4d-a32f-b9d441a207d2
OS Type:     hvm
State:       running
CPU(s):      2
CPU time:    46.7s
Max memory:  4194304 KiB
Used memory: 4194304 KiB
Persistent:  yes
Autostart:   disable
Autostart Once: disable
```

Managed save: no
Security model: apparmor
Security DOI: 0
Security label: libvirt-2488c989-95f1-4b4d-a32f-b9d441a207d2 (enforcing)

loke4884@linuxforge:~/kvm-vms/linuxserver2\$ sudo virsh net-dhcp-leases default

Expiry Time	MAC address	Protocol	IP address	Hostname	Client ID or DUID
-------------	-------------	----------	------------	----------	----------------------

2026-09-07 12:35:36	52:54:00:d8:7b:84	ipv4	192.168.122.31/24	linuxserver2	ff:56:50:4d:98:00:02:00:00:ab:11:aa:9b:03:64:1f:1e:52:a1
---------------------	-------------------	------	-------------------	--------------	--

loke4884@linuxforge:~/kvm-vms/linuxserver2\$ ssh owner@192.168.122.31

The authenticity of host '192.168.122.31 (192.168.122.31)' can't be established.

ED25519 key fingerprint is: SHA256:qZZ7xmZe2v+NjwiTne3rDV0vc5ly21eTZ5rAFh/THQ8

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added '192.168.122.31' (ED25519) to the list of known hosts.

owner@192.168.122.31's password:

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Mon Sep 7 06:11:34 UTC 2026

System load: 0.01	Processes: 134
Usage of /: 10.1% of 18.33GB	Users logged in: 1
Memory usage: 5%	IPv4 address for enp1s0: 192.168.122.31
Swap usage: 0%	

Expanded Security Maintenance for Applications is not enabled.

68 updates can be applied immediately.

68 of these updates are standard security updates.

To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: sudo pro status

To run a command as administrator (user "root"), use "sudo <command>".

See "man sudo_root" for details.

